

Multinational Firms and Agricultural Trade in Latin America

Ownership and Exports in HS 01–24

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This version: September 2026

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Abstract

We match the customs export records of nine Latin American countries for 2006–2022 to the ownership data of Orbis and Dun & Bradstreet, and use them to describe who owns the region’s agricultural exporters. Agriculture here is HS chapters 01–24 and is worth US\$78.9 billion a year across the nine countries. Multinational affiliates sell 48% of it, 38% foreign owned and 10% domestically owned, with shares that run from 29% in El Salvador to 66% in Paraguay. The owners are concentrated in a handful of countries, most of them grain traders and the holding jurisdictions they use. Foreign presence is heaviest in oil seeds, cereals and animal feed and lightest in sugar and beverages. Two of the patterns that hold for the region’s exports as a whole do not hold inside agriculture: foreign affiliates are no more concentrated in complex or differentiated goods than in simple ones, and the product range is too narrow for a complexity gradient to appear. The other four hold. A few large groups carry the value, cells with more multinationals trade more, and distance holds multinational affiliates back less than other firms.

Keywords: Multinational Firms, Agricultural Trade, Ownership, Firm-Level Data, Latin America
JEL codes: F14, F23, Q17, L25

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1 Introduction

Customs records show what a country sells abroad and to whom. They do not show who owns the seller. In agriculture that gap is wide, because a large part of what Latin America ships is handled by grain traders, processors and food companies headquartered elsewhere, and the aggregate investment statistics do not say which firm ships which crop to which market.

This paper closes the gap for nine countries: Argentina, Chile, Colombia, Costa Rica, the Dominican Republic, El Salvador, Paraguay, Peru and Uruguay. We match their customs export records for 2006–2022 to the ownership records of Orbis and Dun & Bradstreet, label every exporter as a foreign affiliate, a domestic affiliate or a local firm, and restrict the result to agriculture, defined as HS chapters 01–24. That is US\$78.9 billion a year, a third of what the nine countries export. Everything here is for the region as a whole.

Five results are worth stating up front. Multinational affiliates sell 48% of the region’s agricultural exports, 38% of it foreign owned, and the share ranges from 29% in El Salvador to 66% in Paraguay. Ownership is concentrated in a few countries, most of them grain traders and the holding jurisdictions they use. Within agriculture the foreign share is highest in oil seeds (62%), miscellaneous edible products (71%) and tobacco (55%), and lowest in sugar (19%) and beverages (29%). Groups with more than a hundred affiliates worldwide are 12% of parents and half of foreign-affiliate value, while 45% of parents have a single affiliate. And the two regularities that hold for merchandise exports as a whole do not survive inside agriculture: there is no complexity gradient and no difference between differentiated and homogeneous goods.

Contribution to the literature: to be written.

Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the facts within agriculture. Section 5 looks inside the sector. Section 6 sketches the model and Section 7 concludes.

2 Related Literature

To be written. Three strands: firm-level evidence on multinational production and trade; ownership, market power and trade policy; and descriptive evidence on multinational firms in Latin America.

3 Data

3.1 Customs transactions

The trade data are the export records of the customs authorities of nine Latin American countries, obtained by the Inter-American Development Bank under its agreements with the national agencies. A record gives the value exported (FOB, current US dollars) by one identified firm, of one six-digit HS 2007 product, to one destination, in one year. Firms are identified by their national tax number, which is unique within a country, and by the name written on the declaration.

Coverage is the universe of merchandise exports in the years each country made available. Those years differ, from Chile’s 2009–2022 to Paraguay’s 2012–2020, and all nine overlap between 2012 and 2018. Because the panel is unbalanced, every dollar figure in this paper is an annual average, the country’s pooled value divided by the number of years it is observed, while shares are computed on pooled values. Ecuador is in the raw data but excluded throughout. Services do not appear in customs merchandise records and are outside the paper.

Agriculture is HS chapters 01–24, which covers live animals, meat, fish, dairy, vegetables, fruit, cereals, oil seeds, fats and oils, prepared foods, beverages and tobacco. Every figure in this paper is

computed on that subset of the records described above, for the nine countries pooled. We do not report country-specific or partner-specific analyses.

3.2 Ownership records

Ownership comes from two commercial databases. Orbis (Bureau van Dijk) gives the ownership links between companies: for each subsidiary, the identity and country of its global ultimate owner at the 25% control threshold. Dun & Bradstreet gives an independent roster with the DUNS number of each establishment and of its global ultimate, together with names, countries and industries. We merge the two into one file in which each firm carries a group identifier, the parent's country and name, and the number of affiliates the group has worldwide, which is the measure of network size behind Fact 4. The two sources overlap without coinciding: Orbis covers the ownership chain better, Dun & Bradstreet the operating establishments, and some exporters appear in only one.

3.3 Matching exporters to their owners

Customs gives a tax number and a name typed into a declaration. The corporate databases give a registered company name and, for some countries, the same tax number. Matching runs in four steps.

1. **Tax identifier.** Direct match on the national tax number wherever the corporate database records it. This step delivers about 60% of matched export value.
2. **Fuzzy name matching, country by country.** For the rest, company names are compared with a character-level TF-IDF trigram similarity. Character sequences survive the abbreviations that customs declarations make to company names; words do not. A large language model then accepts or rejects each candidate pair.
3. **Fuzzy name matching, pooled.** A second pass over the firms still unmatched, this time against the corporate databases of all countries at once, which catches affiliates registered under a group name rather than a local one.
4. **Manual review.** For the 500 largest unmatched exporters of each country, the match was checked by hand against company websites, corporate registries and public filings. These firms are few and carry a large share of export value, so they drive the aggregate shares.

A parallel step recovers the parent's country when the ownership record names the parent but not its location, first from the raw Orbis links and then from a model-assisted review of the parent's name and registered address. After it, the country of the ultimate owner is known for 92.5% of what foreign affiliates export.

3.4 Who counts as a multinational

An exporter found in Orbis or Dun & Bradstreet is an affiliate of a multinational group. We split matched exporters by where the group's ultimate owner sits: a **domestic** affiliate when the owner is in the exporting country, a **foreign** affiliate otherwise. Matched exporters whose parent country is unknown count as foreign, so the foreign category is what is left of matched firms after the domestic ones. An exporter found in neither database is a **local** firm.

Two consequences matter for reading the exhibits. Because the foreign category is a residual, the aggregate foreign share does not depend on how many parent countries were recovered, which is what Facts 1, 2, 5 and 6 need. And wherever an exhibit splits the foreign bar *by* parent country, the 7.5% of foreign-affiliate value with an unknown parent is allocated across parent countries in proportion to the recorded parents of the same group, so those shares still refer to all foreign affiliates and still add up with the domestic and local ones.

One limit is worth stating. The match runs through ownership links, so a group’s head company cannot be identified as an exporter: every matched exporter in the data has an ultimate owner different from itself, and a Latin American group’s head company exporting from home falls among the unmatched. What the data do show is whether the group is present *at the destination*, either because the affiliate ships to its parent’s country or because the group has another affiliate there. Facts 5 and 6 use that distinction.

3.5 Product classifications and other sources

Products carry the Hausmann–Hidalgo Product Complexity Index, the four technology categories of Lall (2000), the conservative Rauch (1999) split into differentiated, reference-priced and exchange-traded goods, BEC Rev. 4 end use and categories, SITC Rev. 3 divisions, NAICS codes, upstreamness, and the import-demand elasticities of Fontagné, Guimbard and Orefice (2022), all mapped to HS 2007 six-digit lines. Sectors are defined on HS chapters: agriculture 01–24, mining and fuels 25–27 and 71, manufacturing 28–97 excluding 71. Distance, contiguity, common language and trade agreements come from the CEPII gravity database.

3.6 The working sample

Table 1 describes the agricultural sample country by country and Table 2 summarizes it. Regressions use plain logarithms, so a cell with a zero count drops out of that column, and cluster standard errors at the origin–destination level.

Table 1: The agricultural sample: coverage by exporting country

Origin	Years	Exporters	Matched	HS6 lines	Destinations	Records	Value (\$bn/yr)
ARG	2011–2019	6,709	1,940	686	195	217,027	38.1
PER	2010–2019	7,366	1,219	658	193	163,296	8.1
CHL	2009–2022	5,015	1,833	783	196	245,797	7.5
COL	2010–2021	7,816	1,458	718	206	164,576	7.2
PRY	2012–2020	822	252	272	155	16,893	6.2
URY	2010–2019	1,269	355	510	188	34,132	4.4
CRI	2010–2019	3,776	362	700	174	80,079	4.3
DOM	2012–2019	4,677	434	601	172	95,340	2.0
SLV	2006–2018	1,382	140	610	125	33,701	1.1
All nine	2006–2022	38,832	7,993	1,023	245	1,050,841	78.9

One record = firm $\times$ destination $\times$ HS6 product $\times$ year. Exporters are counted as distinct tax identifiers within an origin; matched exporters are those found in Orbis or Dun & Bradstreet after the tax-identifier match, the fuzzy name match and the manual review of the largest exporters. Value is the annual average: the origin’s pooled export value divided by the number of years it is observed. Ecuador is in the raw data but is excluded throughout.

Table 2: The agricultural sample in figures

The working sample	
Exporting countries	9
Years	2006–2022
Records (firm $\times$ destination $\times$ HS6 $\times$ year)	1,050,841
Exporters	38,832
matched to Orbis / Dun & Bradstreet	7,993
HS6 product lines	1,023
Destination markets	245
Parent countries of foreign multinationals	130
Multinational groups	4,584
Export value, pooled (\$bn)	788
foreign multinationals	38.1%
domestic multinationals	9.5%
local firms	52.4%
Foreign-MNE value with a recorded parent country	87.7%

4 The Six Facts within Agriculture

4.1 Fact 1: Multinational corporations account for a large share of agricultural export values

Figure 1 gives, for each country, the share of agricultural export value sold by multinational affiliates. Multinationals sell 48% of the region's agricultural exports, 38% foreign owned and 10% domestically owned. The spread across countries is wide: 66% in Paraguay and 62% in Uruguay at one end, 29% in El Salvador and 37% in Peru at the other. Foreign owners are ahead of domestic ones everywhere, and the gap is narrowest in Chile (30% against 19%), where fruit and wine have domestically owned groups of some size.

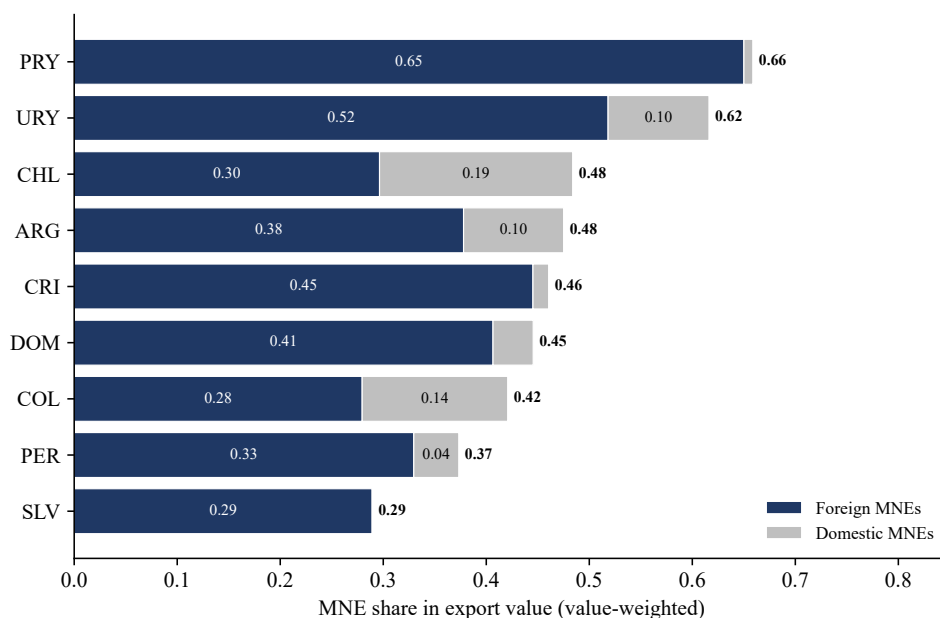
**Figure 1:** Agriculture: MNE share of export value by origin, foreign and domestic multinationals.

Table 3: Numbers behind the previous figure

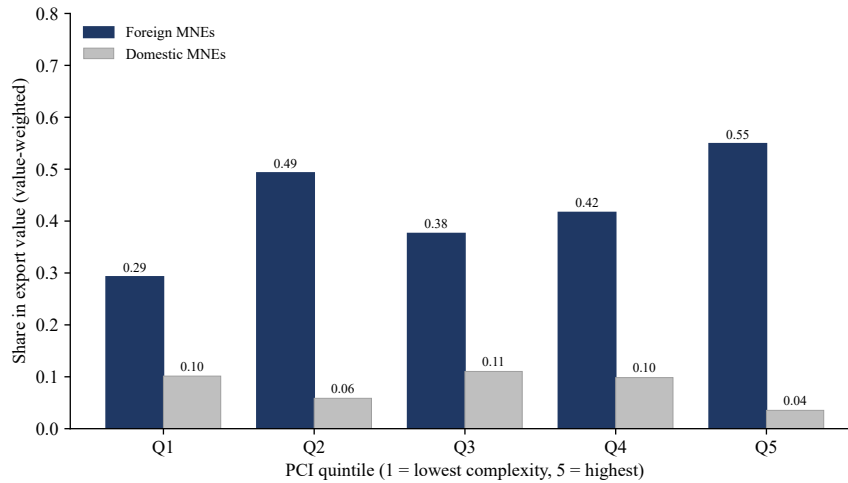
Origin	Foreign	Domestic	MNE total	Value (\$bn/yr)
PRY	0.650	0.009	0.659	6.2
URY	0.518	0.098	0.616	4.4
CHL	0.296	0.187	0.484	7.5
ARG	0.378	0.097	0.475	38.1
CRI	0.445	0.015	0.461	4.3
DOM	0.407	0.039	0.446	2.0
COL	0.280	0.141	0.421	7.2
PER	0.330	0.044	0.373	8.1
SLV	0.289	0.000	0.289	1.1
All	0.381	0.095	0.476	78.9

Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

4.2 Fact 2: Inside agriculture, sophistication does not sort owners

This is where agriculture parts company with the rest of the region's exports. Across merchandise exports as a whole the foreign share rises with product complexity and is higher in differentiated goods. Inside agriculture neither holds. Across quintiles of the Product Complexity Index the foreign share reads 0.29, 0.49, 0.38, 0.42, 0.55, so the two ends go the right way and the middle does not (Figure 2). By Rauch class the foreign share is 41% in differentiated goods, 32% in reference-priced ones and 41% in exchange-traded ones (Table 4 and Figure 3). The domestic share sits near 9% in all three. By end use the foreign share is 41% in intermediates against 35% in consumption goods (Table 5), a small difference next to the 65% against 44% found in manufacturing. Agricultural products occupy a narrow band of the complexity range, and inside that band sophistication does not sort owners.

Panel A. By quintile of the Product Complexity Index (Q1 = least complex)



Panel B. By Lall (2000) technology category

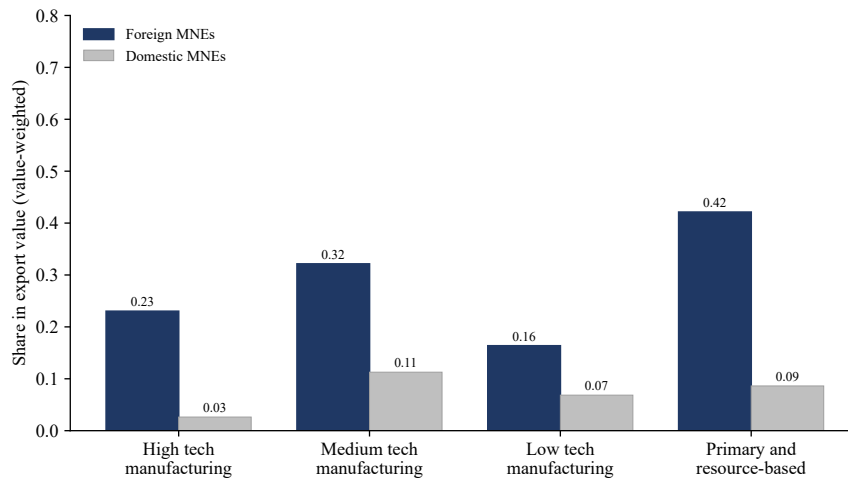


Figure 2: Agriculture: MNE share of export value by product sophistication.

Table 4: Agriculture: MNE shares by Rauch (1999) class

Rauch (1999) class	Foreign	Domestic	MNE total	N HS6	Value (\$bn/yr)
Differentiated	0.413	0.093	0.506	167	7.8
Reference-priced	0.315	0.098	0.413	357	20.1
Homogeneous (exchange)	0.409	0.094	0.503	164	43.7

Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

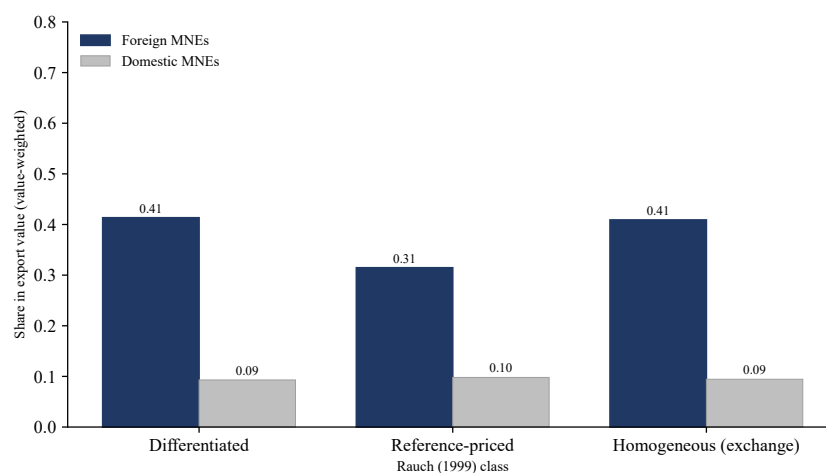


Figure 3: Agriculture: foreign vs domestic MNE shares of export value by Rauch (1999) class.

Table 5: Agriculture: MNE shares by BEC end use

BEC end use	Foreign	Domestic	MNE total	N HS6	Value (\$bn/yr)
Intermediate	0.411	0.090	0.501	243	42.5
Consumption	0.351	0.099	0.450	456	30.4
Capital	0.168	0.020	0.187	3	0.0

BEC Rev. 4 end use in three classes: intermediate = industrial supplies, fuels (incl. motor spirit and goods n.e.s., which are fuels in these data), parts and food for industry; consumption = food for households, passenger cars and consumer goods; capital = capital goods and industrial transport equipment. HS6 lines without a BEC correspondence are excluded. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 6: Agriculture: product sophistication and MNE shares, each measure in its own regression

	PCI		Lall		Rauch	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: MNE_{total} share</i>						
Complexity (PCI)	0.0733*** (0.0061)	0.0614*** (0.0064)				
Lall: high tech (vs primary and resource-based)			-0.2016*** (0.0104)	-0.2103*** (0.0102)		
Lall: medium tech			-0.0395*** (0.0060)	-0.0465*** (0.0066)		
Lall: low tech			-0.2467*** (0.0278)	-0.2272*** (0.0268)		
Rauch: differentiated (vs homogeneous)					0.0435*** (0.0088)	0.0488*** (0.0098)
Rauch: reference-priced					-0.0460*** (0.0068)	-0.0393*** (0.0066)
Observations	158,616	156,478	274,948	272,985	271,105	269,190
<i>Panel B: MNE_{ext} share</i>						
Complexity (PCI)	0.0562*** (0.0067)	0.0490*** (0.0072)				
Lall: high tech (vs primary and resource-based)			-0.1480*** (0.0091)	-0.1576*** (0.0094)		
Lall: medium tech			-0.0623*** (0.0067)	-0.0678*** (0.0073)		
Lall: low tech			-0.2301*** (0.0153)	-0.2262*** (0.0139)		
Rauch: differentiated (vs homogeneous)					0.0554*** (0.0097)	0.0560*** (0.0107)
Rauch: reference-priced					-0.0439*** (0.0069)	-0.0383*** (0.0066)
Observations	158,616	156,478	274,948	272,985	271,105	269,190
<i>Panel C: MNE_{dom} share</i>						
Complexity (PCI)	0.0172*** (0.0031)	0.0123*** (0.0033)				
Lall: high tech (vs primary and resource-based)			-0.0536*** (0.0051)	-0.0527*** (0.0053)		
Lall: medium tech			0.0229*** (0.0032)	0.0212*** (0.0033)		
Lall: low tech			-0.0165 (0.0247)	-0.0010 (0.0273)		
Rauch: differentiated (vs homogeneous)					-0.0120*** (0.0044)	-0.0071* (0.0043)
Rauch: reference-priced					-0.0021 (0.0038)	-0.0010 (0.0040)
Observations	158,616	156,478	274,948	272,985	271,105	269,190
Origin × year and destination × year FE	✓		✓		✓	
Origin × dest. × year FE		✓		✓		✓

One observation = origin *imes* destination *imes* HS6 product *imes* year; dep. var. MNE share of the export value of the product from the origin to the destination in the year; each measure entered in its own regression; weighted by that export value; robust (HC1) SE. PCI = Hausmann–Hidalgo Product Complexity Index (continuous); Lall (2000) in the four categories of Figure 3, base = primary and resource-based; Rauch (1999) in three classes, base = homogeneous (exchange-traded). Cells whose HS6 line lacks the measure are dropped from that column. *** p<0.01, ** p<0.05, * p<0.1

4.3 Fact 3: Ownership is concentrated in a few countries

Figure 4 ranks the owners of foreign affiliates in agriculture by their share of foreign-affiliate export value. Ownership is concentrated: the ten largest owner countries account for roughly three quarters of it. The list is a list of the grain trade and the jurisdictions it holds assets through, with Switzerland (11%), the Netherlands (9.5%) and Liechtenstein (7.6%) high in the ranking next to the United States (24%). Where each owner's affiliates sell is in Tables A.3 and A.4 in the appendix.

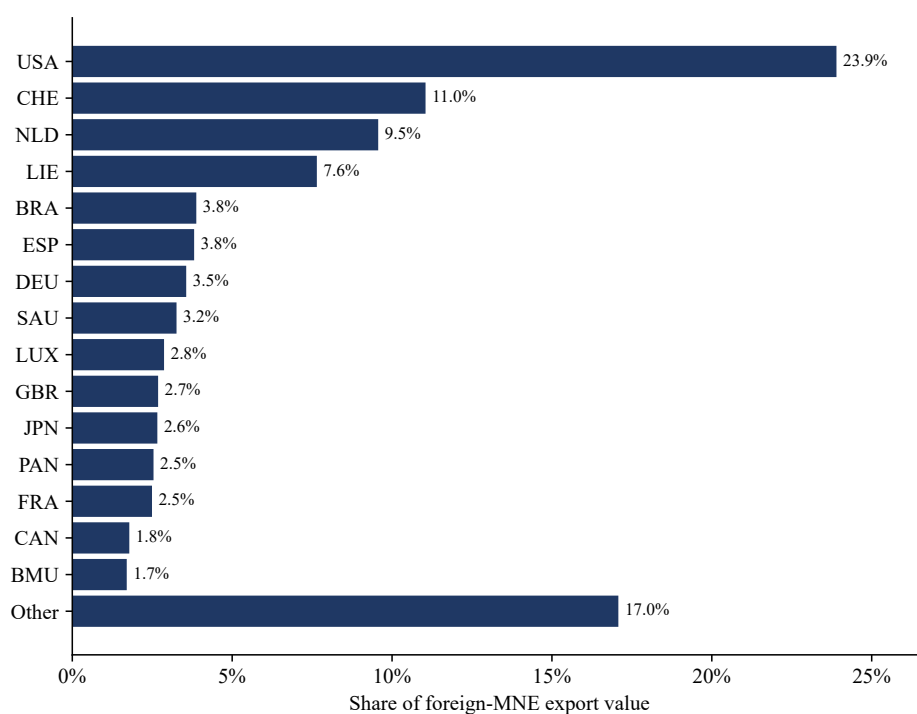


Figure 4: Agriculture: foreign-MNE export value by the parent's country, top 15 and other.

Table 7: Numbers behind the previous figure

Parent country	Share (%)	Value (\$bn/yr)
USA	23.9	6.7
CHE	11.0	3.1
NLD	9.5	2.7
LIE	7.6	2.2
BRA	3.8	1.1
ESP	3.8	1.0
DEU	3.5	0.8
SAU	3.2	0.9
LUX	2.8	0.8
GBR	2.7	0.7
JPN	2.6	0.6
PAN	2.5	0.6
FRA	2.5	0.7
CAN	1.8	0.4
BMU	1.7	0.5
Other (115 countries)	17.0	4.4
All foreign MNEs	100.0	30.7

Shares of foreign-MNE export value by parent country (Figure 4 of the July note). Foreign MNEs whose parent country is not recorded (7.5% of foreign-MNE export value) are allocated to parent countries in proportion to the recorded parents of the same group, so every share by parent country refers to all foreign MNEs and adds up with the domestic and local shares. Value column: recorded parents only, annual averages; the last row is all foreign MNEs. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

4.4 Fact 4: A small set of large groups accounts for the bulk of exports

Figure 5 groups the owners of foreign affiliates by the size of their worldwide network. Agriculture has a long tail of small owners: 45% of them have a single affiliate, a larger share than in the region's exports as a whole. The value still sits with the large groups, with owners of more than a hundred affiliates accounting for 12% of owners and 50% of foreign-affiliate export value.

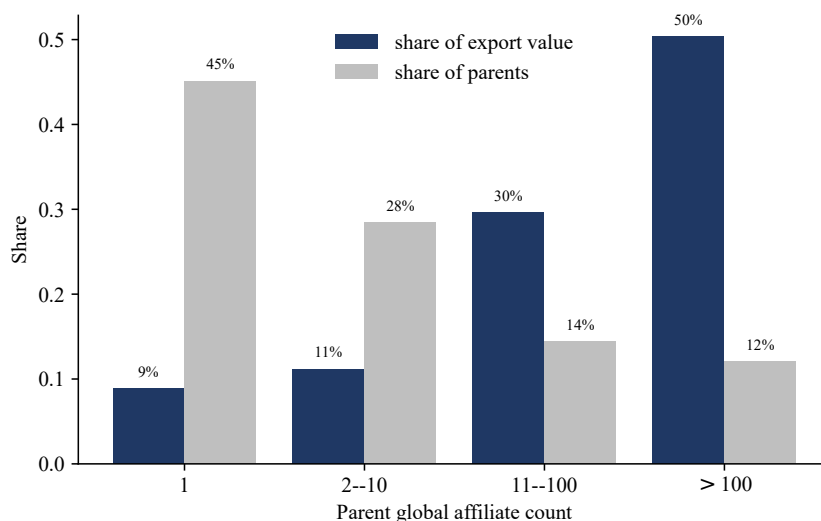


Figure 5: Agriculture: foreign-MNE export value and number of parents by size of the group's global affiliate network.

Table 8: Numbers behind the previous figure

Global affiliate count	Parents	Share of parents (%)	Share of foreign-MNE export value (%)
1	1,436	45.1	8.9
2-10	904	28.4	11.2
11-100	459	14.4	29.6
>100	383	12.0	50.3
All parents with a network record	3,182	100	100

Foreign-MNE parents (parent name key) exporting from the nine origins; 98% of parents and 99% of foreign-MNE export value have a matched global-network record (Orbis $\cup$ D&B roster, src/16). Network size = number of distinct worldwide affiliates.

4.5 Fact 5: Greater multinational presence is associated with higher trade volumes

Table 9 relates the agricultural exports of an origin, destination, product and year cell to the multinational presence in it. The associations are the same as for merchandise exports as a whole, and they hold for the exports of the non-multinational firms in the same cell (Panel B). Table 10 splits the intensive margin by owner: 1.30 for foreign affiliates against 0.52 for domestic ones, the same factor of roughly three found in the region's exports as a whole. The extensive margin is in Table 11.

Table 9: Agriculture: multinational presence and trade volume, intensive and extensive margins

	Intensive: ln(# MNE firms)			Extensive: any MNE present		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A: all exports (ln)</i>						
ln(# MNE firms)	2.1821*** (0.0283)	2.1861*** (0.0279)	0.8509*** (0.0190)			
Any MNE present				1.7857*** (0.0611)	1.7863*** (0.0604)	0.6860*** (0.0232)
<i>Panel B: non-MNE exports (ln)</i>						
ln(# MNE firms)	1.5538*** (0.0332)	1.5628*** (0.0332)	0.4048*** (0.0245)			
Any MNE present				1.9365*** (0.0567)	1.9313*** (0.0554)	0.2812*** (0.0181)
Origin FE	✓			✓		
Destination FE	✓			✓		
Product FE	✓	✓		✓	✓	
Year FE	✓			✓		
Origin × year FE		✓			✓	
Destination × year FE		✓			✓	
Origin × dest. × product FE			✓			✓
Origin × dest. × year FE			✓			✓
Observations (all exports)	154,199	153,847	137,324	296,092	295,788	268,332
Observations (non-MNE exports)	91,020	90,655	81,670	232,907	232,617	208,479

Origin-destination-product-year cells; dep. var. ln exports. Intensive margin: ln number of MNE firms (cells with ≥ 1 MNE); extensive margin: indicator for any MNE present. SE clustered at origin-destination. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 10: Agriculture, intensive margin: ln(# MNE firms) decomposed into foreign / domestic

	All MNEs		Foreign / domestic		Foreign through HQ / not through HQ / domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
ln(# MNE firms)	2.1861*** (0.0279)	0.8509*** (0.0190)				
ln(# foreign MNEs)			1.2954*** (0.0275)	0.4625*** (0.0188)		
ln(# foreign MNEs present through HQ)					0.4928*** (0.1145)	0.2334*** (0.0804)
ln(# foreign MNEs not present through HQ)					0.9569*** (0.0754)	0.3804*** (0.0360)
ln(# domestic MNEs)			0.5233*** (0.0266)	0.2831*** (0.0234)	0.4096*** (0.0753)	0.1935*** (0.0421)
<i>Panel B: non-MNE exports (ln)</i>						
ln(# MNE firms)	1.5628*** (0.0332)	0.4048*** (0.0245)				
ln(# foreign MNEs)			1.2191*** (0.0363)	0.3602*** (0.0292)		
ln(# foreign MNEs present through HQ)					0.3864*** (0.1357)	0.2698** (0.1255)
ln(# foreign MNEs not present through HQ)					1.1003*** (0.1197)	0.4395*** (0.0693)
ln(# domestic MNEs)			0.5676*** (0.0339)	0.2461*** (0.0341)	0.4367*** (0.1236)	0.2032** (0.0759)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (all exports)	153,847	137,324	25,437	22,002	1,952	1,604
Observations (non-MNE exports)	90,655	81,670	22,503	19,535	1,903	1,567

Origin-destination-product-year cells; dep. var. ln exports. Intensive margin: plain logs of the counts, so each column uses the cells where every count entering it is positive (no $\ln(1+x)$). (1)–(2) all MNEs; (3)–(4) foreign vs domestic; (5)–(6) foreign split by whether the group is present at the destination through its headquarters (the parent's country). SE clustered at origin-destination. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 11: Agriculture, extensive margin: presence indicators decomposed into foreign / domestic

	All MNEs		Foreign / domestic		Foreign through HQ / not through HQ / domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: all exports (ln)</i>						
Any MNE present	1.7863*** (0.0604)	0.6860*** (0.0232)				
Any foreign MNE			1.7972*** (0.0558)	0.6929*** (0.0213)		
Any foreign MNE present through HQ					1.6277*** (0.1122)	0.5045*** (0.0603)
Any foreign MNE not present through HQ					1.7886*** (0.0563)	0.6904*** (0.0217)
Any domestic MNE			1.6133*** (0.0577)	0.5683*** (0.0237)	1.5621*** (0.0556)	0.5650*** (0.0238)
<i>Panel B: non-MNE exports (ln)</i>						
Any MNE present	1.9313*** (0.0554)	0.2812*** (0.0181)				
Any foreign MNE			1.7649*** (0.0546)	0.2854*** (0.0179)		
Any foreign MNE present through HQ					1.4262*** (0.1036)	0.2467*** (0.0420)
Any foreign MNE not present through HQ					1.7354*** (0.0537)	0.2888*** (0.0179)
Any domestic MNE			1.3082*** (0.0539)	0.2610*** (0.0201)	1.2541*** (0.0504)	0.2590*** (0.0201)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (all exports)	295,788	268,332	295,788	268,332	295,788	268,332
Observations (non-MNE exports)	232,617	208,479	232,617	208,479	232,617	208,479

One observation = origin *imes* destination *imes* HS6 product *imes* year; dep. var. ln exports of the product from the origin to the destination in the year. Extensive margin: indicators for at least one MNE of each type exporting the product to the destination; all observations. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

4.6 Fact 6: Distance is a weaker barrier to trade for multinational corporations

Table 12 estimates the distance elasticity of firm-level agricultural exports. The baseline elasticity is −0.21, steeper than for merchandise exports as a whole, and Table 13 shows the attenuation for multinationals: 0.037 for foreign affiliates against 0.019 for domestic ones. Distance matters more in agriculture, and multinational affiliates escape less of it than they do elsewhere, but the ordering between foreign and domestic owners is unchanged.

Table 12: Agriculture: distance and firm exports, multinationals split by presence at the destination

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.2118*** (0.0748)	-0.2269*** (0.0712)	-0.2078*** (0.0745)	-0.2224*** (0.0708)
ln distance × MNE	0.0326*** (0.0031)	0.0318*** (0.0031)		
ln distance × MNE, present			0.0890*** (0.0070)	0.0875*** (0.0069)
ln distance × MNE, not present			0.0228*** (0.0032)	0.0221*** (0.0032)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	1,045,009	1,044,787	1,045,009	1,044,787

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. multinationals (foreign or domestic) split by presence at the destination (the group has an affiliate there, or the destination is the parent's country). SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

Table 13: Agriculture: distance and firm exports, foreign / domestic

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.2135*** (0.0748)	-0.2285*** (0.0711)	-0.2130*** (0.0748)	-0.2280*** (0.0711)
ln distance × foreign MNE	0.0366*** (0.0034)	0.0358*** (0.0034)		
ln distance × foreign MNE, present through HQ			0.0610*** (0.0165)	0.0590*** (0.0165)
ln distance × foreign MNE, not present through HQ			0.0359*** (0.0035)	0.0351*** (0.0034)
ln distance × domestic MNE	0.0187*** (0.0046)	0.0182*** (0.0046)	0.0187*** (0.0046)	0.0182*** (0.0046)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	1,045,009	1,044,787	1,045,009	1,044,787

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. (1)–(2) foreign vs domestic multinationals; (3)–(4) foreign MNEs split by whether the group is present at the destination through its headquarters (the parent's country) or not. SE clustered at origin-destination. *** p<0.01, ** p<0.05, * p<0.1

5 Inside Agriculture

Agriculture is not one market. Table 14 splits it into SITC Rev. 3 divisions and Figure 6 shows the same shares. Animal feed is the largest division at US\$13.8 billion a year, followed by vegetables and fruit (10.6), cereals (10.3), oil seeds (6.7) and fixed vegetable oils (6.0). Foreign presence varies by a factor of four across them: 71% in miscellaneous edible products, 62% in oil seeds, 55% in tobacco and 49% in cereals, against 31% in coffee, tea and cocoa, 29% in beverages and 19% in sugar and honey. Domestic multinationals are largest in dairy (20%), coffee (16%) and sugar (15%).

Two other cuts are worth keeping. Table 15 and Figure 7 take the HS sections inside agriculture, which separate live animals and animal products, vegetable products, fats and oils, and prepared food-stuffs. Table 17 and Figure 8 split the sector into agricultural inputs and everything else, with the largest input lines in Table 18. Table 16 adds the BEC end-use split, which separates what leaves the region for further processing from what leaves it ready for a household.

At the product level, Figure 9 gives the twenty largest HS6 lines in agriculture with the foreign, domestic and local shares of each, and Figure 10 splits the foreign bar by owner country.

Table 14: Agriculture by SITC Rev. 3 division: total and MNE exports, shares, leading parents

SITC Rev. 3 division	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
08 Animal feed	13.8	6.6	35.7	11.5	CHE 0.28, LIE 0.19, USA 0.18
05 Vegetables & fruit	10.6	4.0	32.6	5.8	ESP 0.14, USA 0.13, PAN 0.11
04 Cereals	10.3	5.9	48.6	8.2	USA 0.33, CHE 0.14, NLD 0.14
22 Oil seeds	6.7	4.4	61.6	4.1	USA 0.54, NLD 0.14, LIE 0.08
07 Coffee, tea, cocoa, spices	5.1	2.3	30.4	15.6	DEU 0.41, ITA 0.14, SGP 0.12
42 Fixed vegetable oils	6.0	2.5	30.8	10.2	CHE 0.27, NLD 0.24, USA 0.23
01 Meat	5.2	2.4	34.8	13.1	SAU 0.46, BRA 0.26, JPN 0.08
03 Fish & seafood	3.4	1.8	41.9	9.5	ESP 0.32, LUX 0.11, ITA 0.10
29 Crude animal/veg. materials	2.2	0.8	27.3	9.0	DEU 0.10, ARG 0.09, BGR 0.08
11 Beverages	2.2	0.8	28.6	9.2	LUX 0.17, PAN 0.15, GBR 0.12
02 Dairy & eggs	1.9	1.0	28.0	20.3	NLD 0.20, CHE 0.19, ROU 0.19
09 Misc. edible products	1.7	1.2	70.9	2.3	USA 0.46, CHE 0.17, FRA 0.09
06 Sugar & honey	1.4	0.5	19.3	14.7	USA 0.50, KOR 0.18, LIE 0.10
12 Tobacco	1.2	0.7	54.8	0.2	USA 0.36, HKG 0.25, GBR 0.12
41 Animal oils & fats	0.5	0.2	23.0	11.0	BMU 0.31, PAN 0.23, GBR 0.16
Other	0.6	0.1	21.6	1.9	CHE 0.21, USA 0.20, ROU 0.10
All	72.9	35.0	38.5	9.4	

Agriculture (HS 01–24), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

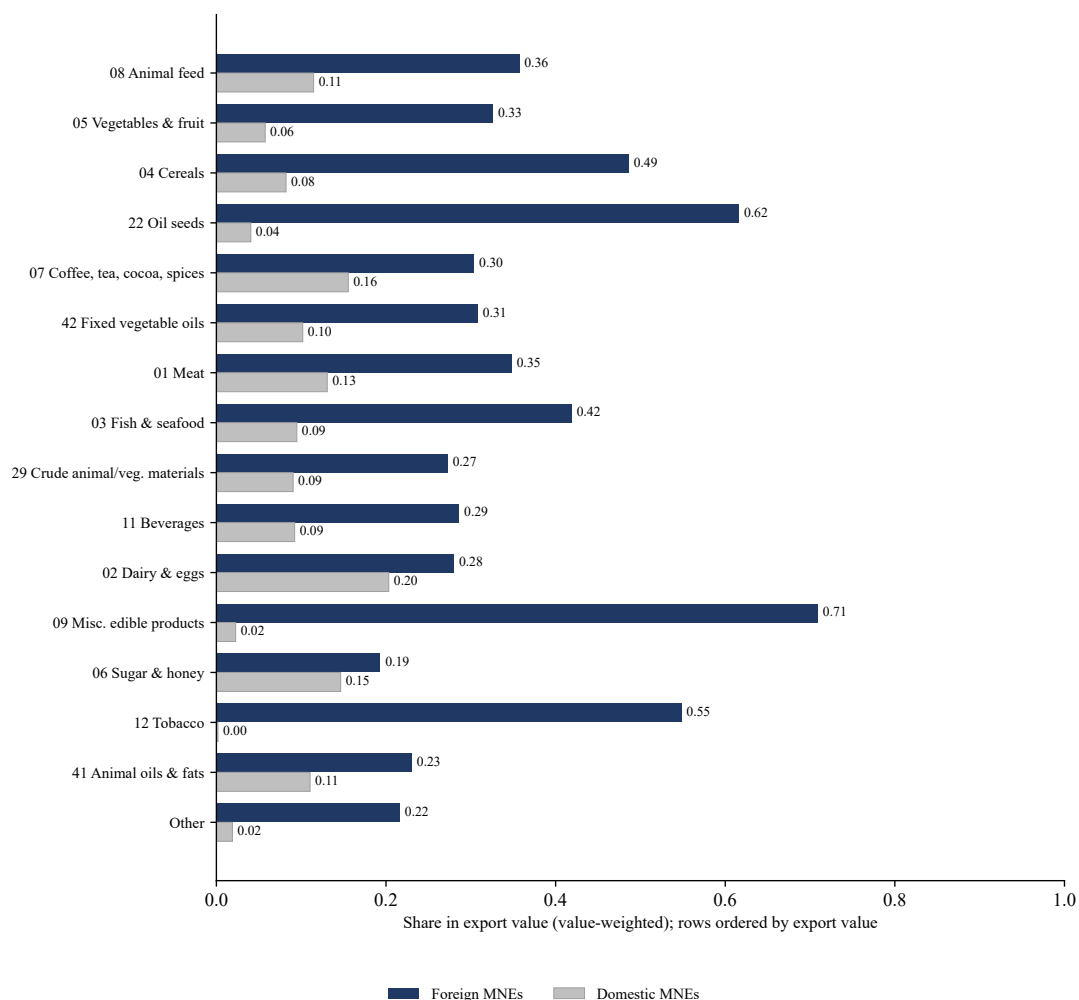
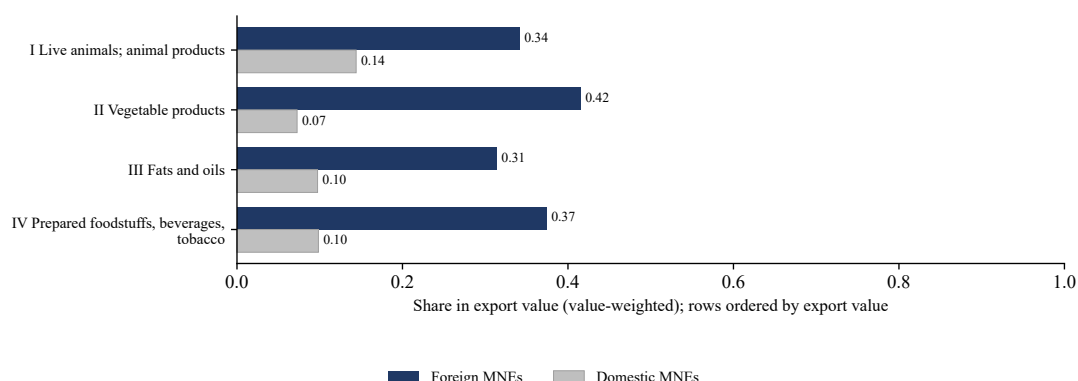
**Figure 6: Agriculture by SITC Rev. 3 division: foreign and domestic MNE shares of export value.**

Table 15: Agriculture by HS section: total and MNE exports, shares, leading parents

HS section	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
I Live animals; animal products	12.5	6.0	34.1	14.4	SAU 0.19, JPN 0.14, ESP 0.10
II Vegetable products	33.5	16.6	41.5	7.3	USA 0.33, NLD 0.11, LIE 0.07
III Fats and oils	6.9	2.9	31.4	9.7	CHE 0.26, NLD 0.21, USA 0.19
IV Prepared foodstuffs, beverages, tobacco	25.9	12.3	37.4	9.8	USA 0.21, CHE 0.17, LIE 0.10
All	78.9	37.8	38.1	9.5	

Agriculture (HS 01–24), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

**Figure 7: Agriculture by HS section: foreign and domestic MNE shares of export value.****Table 16: Agriculture by BEC end use: total and MNE exports, shares, leading parents**

BEC end use	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Intermediate	42.5	21.4	41.1	9.0	USA 0.30, CHE 0.16, NLD 0.13
Consumption	30.4	13.7	35.1	9.9	USA 0.16, ESP 0.09, SAU 0.08
Capital	0.0	0.0	16.8	2.0	GBR 0.35, SAU 0.33, NLD 0.28
All	72.9	35.0	38.5	9.4	

Agriculture (HS 01–24), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

Table 17: Agricultural inputs vs other agricultural goods: total and MNE exports, shares, leading parents

Inputs vs output	Total exports (\$bn/yr)	MNE exports (\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Agricultural inputs	14.9	7.0	36.0	10.8	CHE 0.26, LIE 0.18, USA 0.18
Agricultural output (food, fibres, beverages)	64.0	30.8	38.6	9.2	USA 0.25, NLD 0.09, CHE 0.08
All	78.9	37.8	38.1	9.5	

Agriculture (HS 01–24), pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

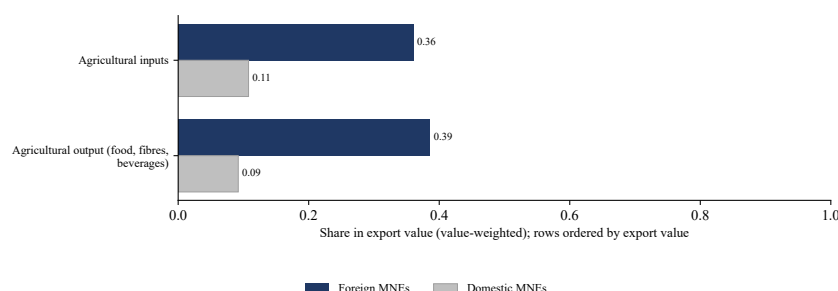


Figure 8: Agricultural inputs vs other agricultural goods: foreign and domestic MNE shares.

Table 18: The largest agricultural-input HS6 lines

HS6	Description	\$bn/yr	Foreign (%)	Domestic (%)
230400	Oil-cake & other solid residues, whether/not ground/in pellets, from extraction of soyabean oil	10.84	35	12
230120	Flours, meals & pellets of fish/of crustaceans, molluscs/other aquatic invertebrates	1.70	31	10
230990	Preparations of a kind used in animal feeding other than dog/cat food put up for RS	0.51	57	3
071333	Kidney beans, incl. white pea beans (<i>Phaseolus vulgaris</i>), dried, shelled, whether/not skinned/split	0.30	28	4
230250	Bran, sharps & other residues, whether/not in the form of pellets, derived from the sifting/milling/other working of leguminous plants	0.28	30	16
100510	Maize (corn), seed	0.19	60	3
230910	Dog/cat food, put up for RS	0.17	81	5
120991	Vegetable seeds, of a kind used for sowing	0.12	67	1
230630	Oil-cake & other solid residues, whether/not ground/in pellets, from extraction of sunflower seeds	0.12	22	2
010290	Live bovine animals other than pure-bred breeding animals	0.08	31	0
071320	Chickpeas (<i>garbanzos</i>), dried, shelled, whether/not skinned/split	0.08	23	2
230800	Vegetable materials/waste/residues/by-products, whether/not in pellets, of a kind used in animal feeding, n.e.s.	0.08	19	0
100610	Rice in the husk (paddy/rough)	0.06	45	5
230110	Flours, meals & pellets of meat/meat offal; greaves	0.05	31	0
051199	Animal products not elsewhere specified/incl. (excl. of 0511.10); dead animals of Ch. 1, unfit for human consumption	0.04	39	5
071310	Peas (<i>Pisum sativum</i>), dried, shelled, whether/not skinned/split	0.03	22	1
071339	Beans (<i>Vigna</i> spp., <i>Phaseolus</i> spp. (excl. of 0713.31-0713.33)), dried, shelled, whether/not skinned/split	0.03	11	2
230230	Bran, sharps & other residues, whether/not in the form of pellets, derived from the sifting/milling/other working of wheat	0.03	16	1
010511	Live fowls of species <i>Gallus domesticus</i> , weighing not >185g	0.02	3	1
120930	Seeds of herbaceous plants cultivated principally for their flowers, of a kind used for sowing	0.02	76	2

Top-20 HS6 lines flagged as agricultural inputs (fertilisers, agrochemicals, seeds for sowing, animal feed, agricultural machinery, live animals); Foreign / Domestic = MNE shares of the line's export value, percent. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

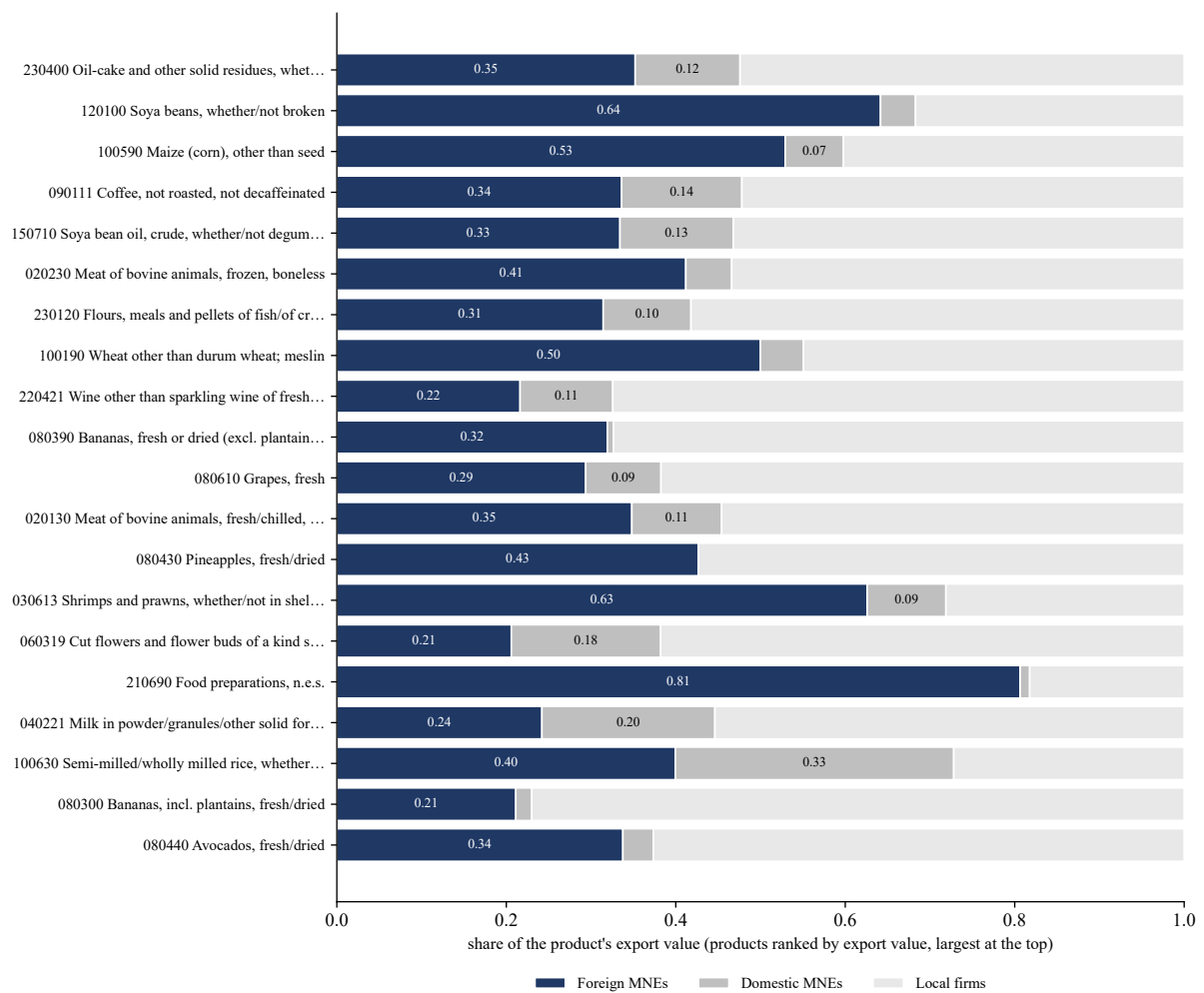


Figure 9: Agriculture, top-20 HS6 products by export value: foreign MNE, domestic MNE and local shares.

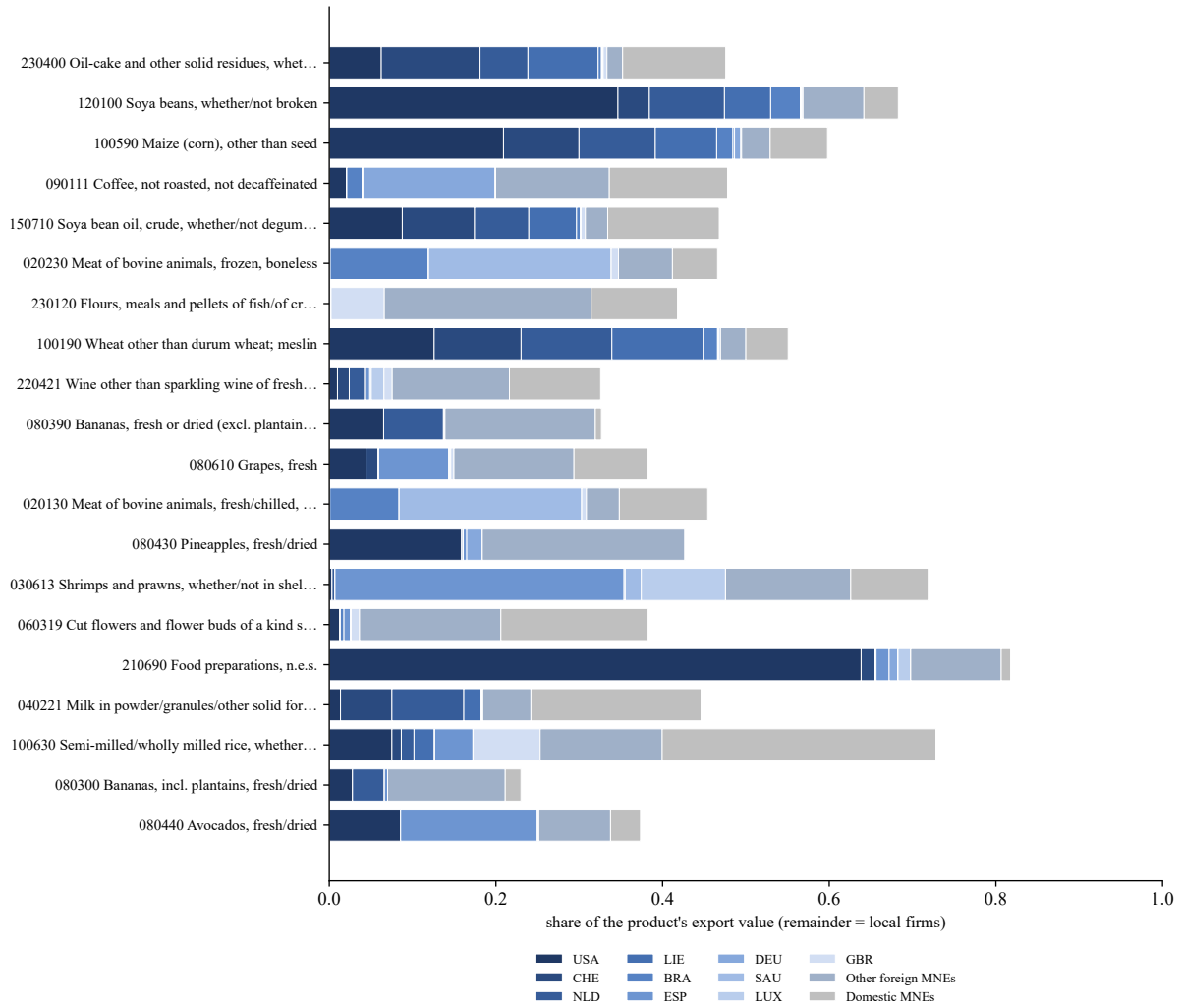


Figure 10: Agriculture, top-20 HS6 products: foreign bar split by the parent's country (remainder = local firms).

6 Theoretical Framework

To be written. The framework is the companion theory paper: an oligopoly model with multinational ownership in which the object of interest is the ownership-weighted Herfindahl rather than the country-level ownership share. The section will state the environment, the objects the facts discipline, and how the model's ownership share maps to the measured one.

7 Concluding Remarks

To be written.

References

To be completed with the literature section. The sources of the classifications used above are:

Fontagné, L., Guimbard, H. and Orefice, G., 2022. Tariff-Based Product-Level Trade Elasticities. *Journal of International Economics*, 137.

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- Lall, S., 2000. The Technological Structure and Performance of Developing Country Manufactured Exports, 1985–98. *Oxford Development Studies*, 28(3).
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A Appendix: Additional Exhibits

This appendix collects the exhibits that document or extend those of the main text, all of them computed within agriculture for the nine origins pooled.

A.1 Fact 1

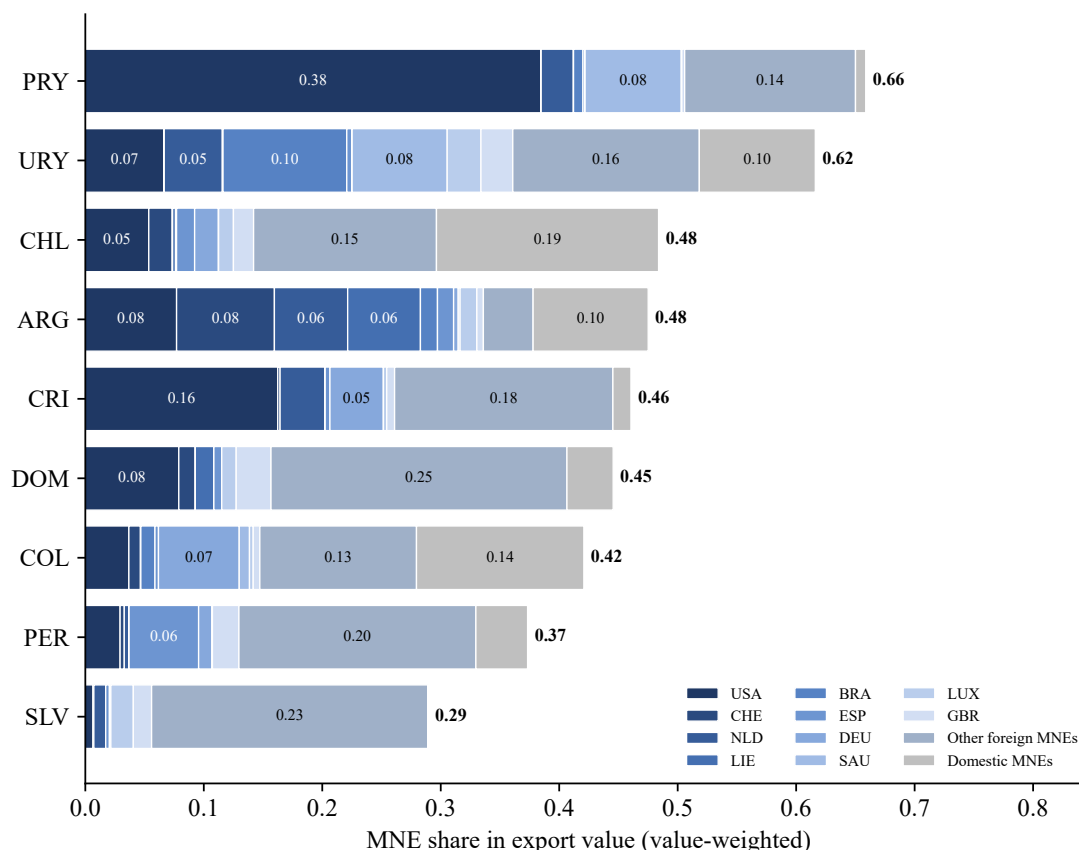


Figure A.1: Agriculture: MNE share of export value by origin, foreign bar split by the parent's country.

Table A.1: Numbers behind the previous figure

Origin	USA	CHE	NLD	LIE	BRA	ESP	DEU	SAU	LUX	GBR	Other	Dom.	MNE total	Value (\$bn/yr)
PRY	0.385	0.000	0.027	0.000	0.008	0.002	0.000	0.081	0.000	0.002	0.144	0.009	0.659	6.2
URY	0.066	0.000	0.049	0.001	0.104	0.004	0.000	0.080	0.028	0.027	0.157	0.098	0.616	4.4
CHL	0.054	0.020	0.000	0.002	0.001	0.015	0.020	0.000	0.012	0.017	0.154	0.187	0.484	7.5
ARG	0.077	0.083	0.062	0.061	0.014	0.014	0.004	0.002	0.014	0.006	0.042	0.097	0.475	38.1
CRI	0.162	0.002	0.038	0.000	0.000	0.004	0.045	0.000	0.003	0.007	0.184	0.015	0.461	4.3
DOM	0.079	0.014	0.000	0.016	0.000	0.007	0.000	0.000	0.012	0.029	0.250	0.039	0.446	2.0
COL	0.037	0.010	0.000	0.000	0.012	0.003	0.068	0.009	0.003	0.006	0.132	0.141	0.421	7.2
PER	0.029	0.004	0.004	0.000	0.000	0.059	0.011	0.000	0.000	0.022	0.200	0.044	0.373	8.1
SLV	0.006	0.001	0.010	0.000	0.000	0.003	0.001	0.000	0.019	0.015	0.233	0.000	0.289	1.1

Figure 1 split by parent country. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and 'Other foreign' (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

A.2 Fact 2

Table A.2: Agriculture: foreign and domestic MNE shares by quintile of each sophistication measure

Measure	Foreign MNE share					Domestic MNE share				
	Q1	Q2	Q3	Q4	Q5	Q1	Q2	Q3	Q4	Q5
PCI	0.29	0.49	0.38	0.42	0.55	0.10	0.06	0.11	0.10	0.04
$ \sigma $ FGO (2022)	0.40	0.40	0.37	0.29	0.33	0.07	0.08	0.12	0.12	0.10
σ Broda–Weinstein	0.36	0.34	0.43	0.47	–	0.12	0.11	0.07	0.05	–
Upstreamness	0.32	0.31	0.37	0.36	0.44	0.08	0.09	0.12	0.11	0.08
Quality ladder	–	–	–	–	–	–	–	–	–	–
RHCI	0.41	0.48	0.37	0.42	0.30	0.10	0.04	0.10	0.09	0.11

Value-weighted shares of export value within quintile; quintiles over HS6 products (unweighted); ‘–’ = fewer than 30 HS6 products carry the measure in this scope. $|\sigma|$ FGO: absolute value of the tariff-based import-demand elasticity of Fontagné, Guimbard and Orefice (2022), HS6 rev. 2007 (non-significant or positive HS6 estimates replaced by the HS4 average by the source); larger = more substitutable. σ Broda–Weinstein, upstreamness (Antràs–Chor), quality ladder (Khandelwal), RHCI (UNCTAD) as in the document’s appendix.

A.3 Fact 3

Table A.3: Agriculture, top-10 parents $\times$ top-10 destinations: destination mix of each parent’s exports (row %)

Parent / Destination	USA	CHN	BRA	ARG	ESP	CHL	JPN	NLD	URY	ITA	Other destinations	Total
USA	7.3	15.5	3.7	16.7	1.2	4.1	1.8	1.7	5.5	0.7	41.8	100.0
CHE	1.7	6.7	7.2	0.2	2.6	3.4	0.4	1.2	0.7	3.7	72.3	100.0
NLD	3.3	14.4	5.1	3.9	1.4	3.6	0.6	2.8	2.4	5.6	57.0	100.0
LIE	1.3	14.4	6.1	0.0	2.9	2.6	0.4	0.9	0.2	2.2	69.1	100.0
BRA	11.4	31.2	4.9	1.9	1.2	1.7	1.1	4.3	0.9	2.2	39.2	100.0
ESP	15.5	8.3	3.0	0.4	26.1	2.5	3.9	5.8	0.2	2.9	31.4	100.0
DEU	37.3	0.6	1.8	0.4	3.5	0.9	5.0	4.3	0.7	1.8	43.6	100.0
SAU	6.4	12.3	6.9	0.1	2.3	15.8	0.1	2.6	0.3	2.1	51.1	100.0
LUX	6.5	2.5	40.8	0.0	8.1	2.0	2.1	0.3	2.8	4.5	30.4	100.0
GBR	8.9	16.9	8.5	0.6	3.5	2.7	2.2	2.9	2.1	0.8	50.9	100.0
Other parents	23.1	7.9	7.9	7.5	4.0	1.5	5.5	3.5	1.8	2.0	35.4	100.0

Row percentages: destination mix of each row group. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational’s ultimate parent as recorded (Orbis/D&B).

Table A.4: Agriculture, top-10 parents $\times$ the same ten countries as destinations (row %)

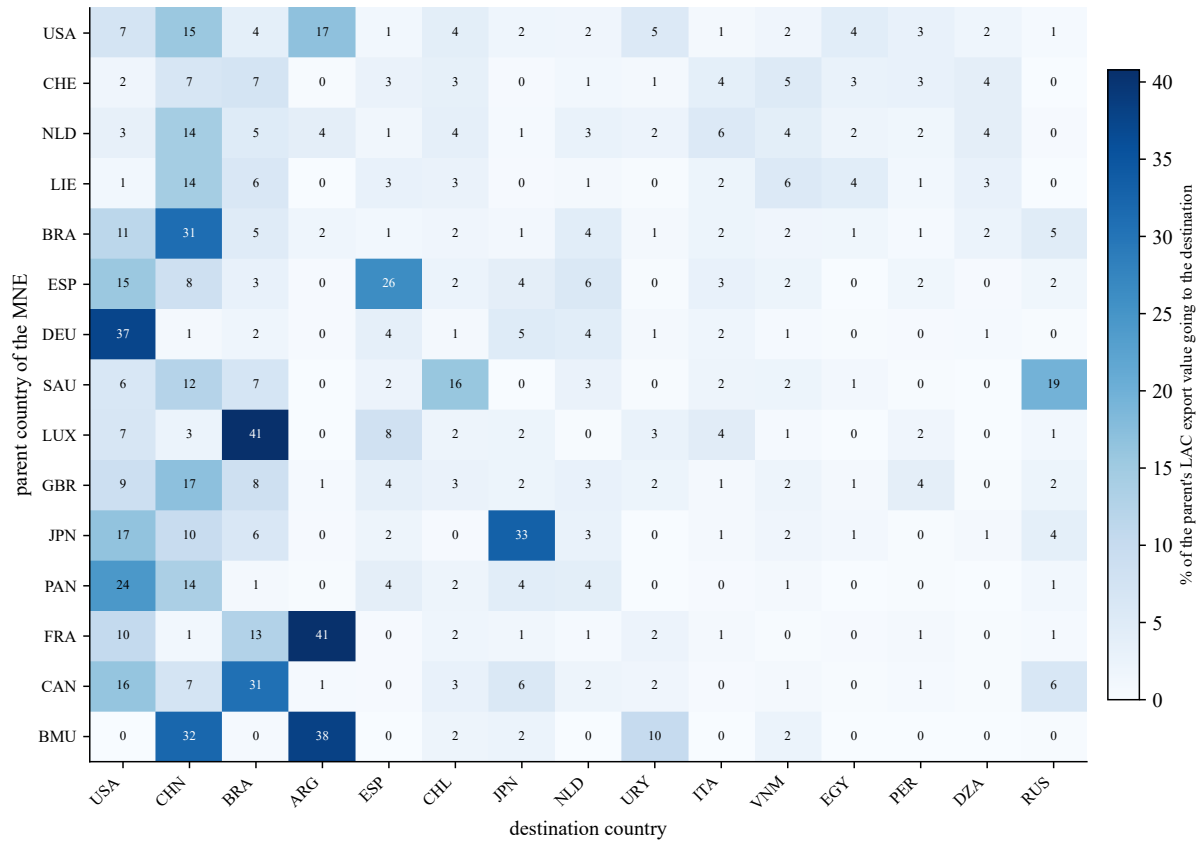
Parent / Destination	USA	CHE	NLD	BRA	ESP	DEU	SAU	LUX	GBR	Other destinations	Total
USA	7.3	0.0	1.7	3.7	1.2	0.6	1.0	0.0	0.5	84.1	100.0
CHE	1.7	0.0	1.2	7.2	2.6	0.3	3.8	0.0	0.9	82.3	100.0
NLD	3.3	0.0	2.8	5.1	1.4	0.6	1.0	0.0	0.8	85.0	100.0
LIE	1.3	0.0	0.9	6.1	2.9	0.1	3.0	0.0	0.5	85.3	100.0
BRA	11.4	0.6	4.3	4.9	1.2	5.1	0.9	0.0	1.3	70.3	100.0
ESP	15.5	0.0	5.8	3.0	26.1	1.6	0.3	0.0	1.9	45.7	100.0
DEU	37.3	0.1	4.3	1.8	3.5	7.1	0.2	0.0	1.2	44.4	100.0
SAU	6.4	0.7	2.6	6.9	2.3	1.2	2.6	0.0	1.5	75.7	100.0
LUX	6.5	0.1	0.3	40.8	8.1	0.7	0.0	0.0	0.4	43.1	100.0
GBR	8.9	0.2	2.9	8.5	3.5	2.1	0.0	0.0	1.6	72.3	100.0
Other parents	23.1	0.3	3.5	7.9	4.0	2.0	0.3	0.0	1.9	56.9	100.0

Row percentages: destination mix of each parent’s exports, with the ten parent countries themselves as the destination columns, so the diagonal is the share shipped to the parent’s own country. Foreign MNEs with a recorded parent country; top 10 parents by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational’s ultimate parent as recorded (Orbis/D&B).

Table A.5: Agriculture, parent region × destination region: destination mix of each group (row %)

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	30.0	17.1	20.5	29.3	2.8	0.3	0.1	100.0
North America	47.9	8.3	10.2	25.6	7.6	0.1	0.4	100.0
Europe	27.3	10.1	22.0	31.6	8.2	0.6	0.2	100.0
Asia	14.2	29.2	24.7	29.1	2.4	0.3	0.2	100.0
Africa	29.5	15.3	30.8	13.1	9.2	0.3	1.8	100.0
Oceania	59.2	21.5	8.5	6.1	0.7	4.1	0.0	100.0
Rest of World	17.8	36.0	28.7	16.8	0.0	0.7	0.0	100.0
Domestic MNEs	20.5	16.0	20.4	34.9	7.0	1.1	0.1	100.0
Local firms (unmatched)	22.8	17.9	26.8	25.6	5.6	1.1	0.2	100.0

Row percentages: destination mix of each row group. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

**Figure A.2:** Agriculture, top-15 parents × top-15 destinations: % of the parent's export value going to each destination.

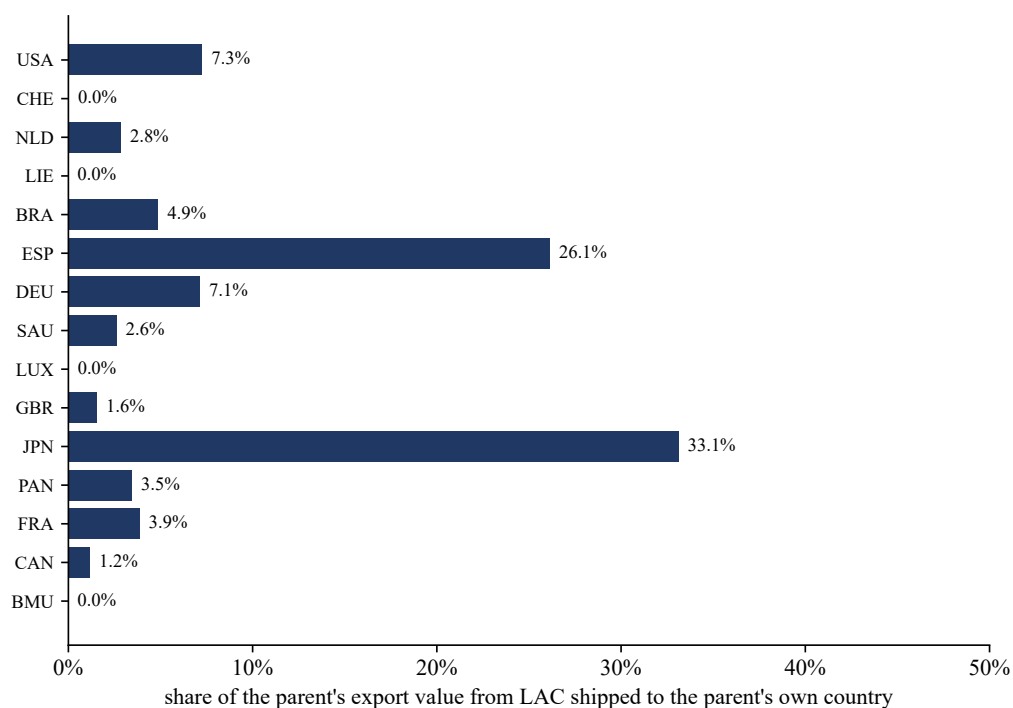


Figure A.3: Agriculture: share of each parent's export value shipped to the parent's own country.

A.4 Fact 5

Table A.6: Agriculture: MNE export value by origin, domestic share and foreign-MNE value by the group's presence at the destination

Origin	MNE value (\$bn/yr)	of which domestic (%)	Foreign-MNE value (\$bn/yr)	% of foreign-MNE value: group present at destination	
				through headquarters	not through HQ (of which via another affiliate)
ARG	18.1	20.4	14.4	2.8	97.2 (30.6)
CHL	3.6	38.7	2.2	12.1	87.9 (29.7)
COL	3.0	33.6	2.0	4.4	95.6 (33.7)
CRI	2.0	3.3	1.9	10.0	90.0 (45.9)
DOM	0.9	8.8	0.8	13.7	86.3 (9.7)
PER	3.0	11.7	2.7	4.7	95.3 (15.5)
PRY	4.1	1.3	4.0	1.9	98.1 (67.8)
SLV	0.3	0.0	0.3	9.5	90.5 (58.9)
URY	2.7	15.9	2.3	2.8	97.2 (42.8)
All	37.8	19.9	30.7	4.7	95.3 (35.8)

'Through headquarters': the affiliate ships to its parent's country. 'Not through HQ': every other destination; in parentheses the part where the group has another affiliate in the destination (Orbis/D&B roster). Pooled 2006–2022. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

A.5 Inside agriculture

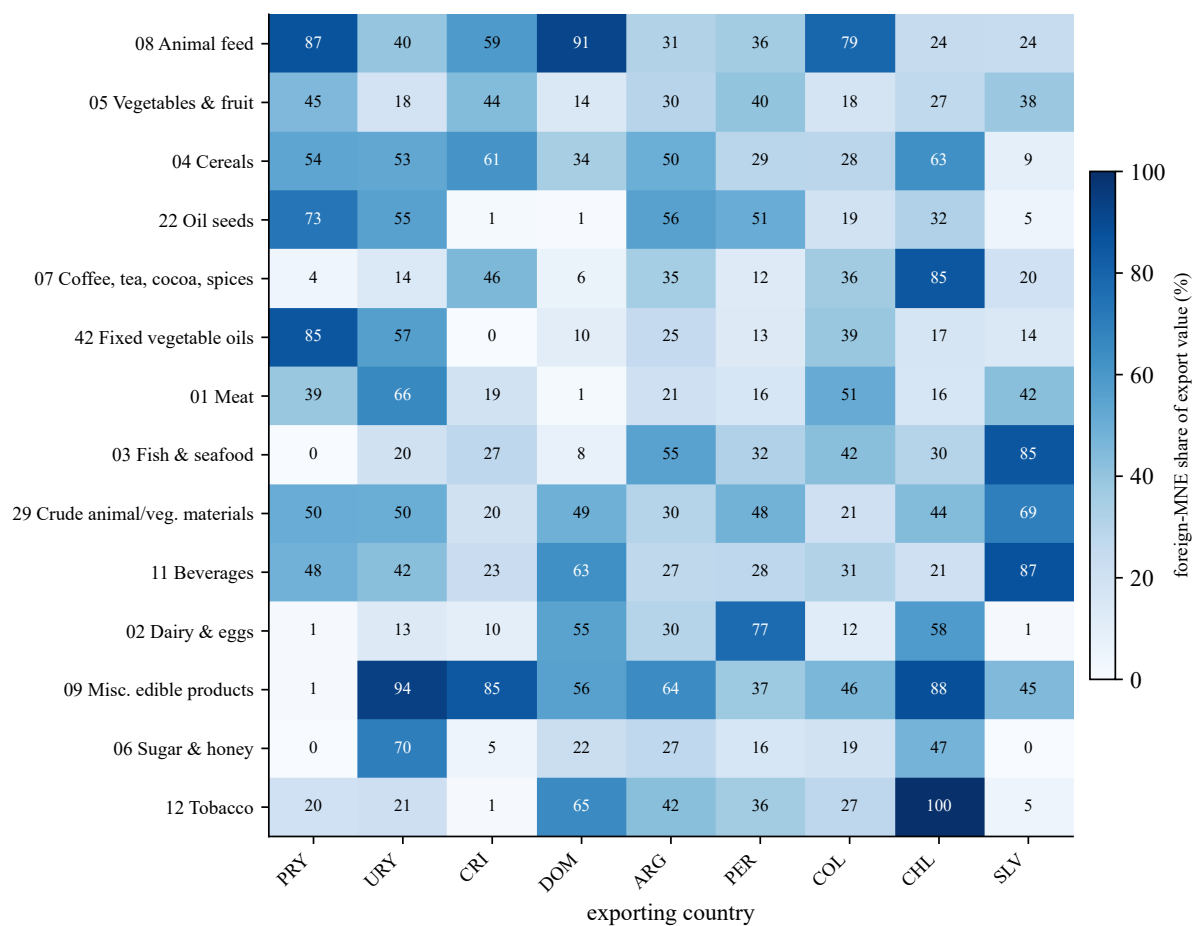


Figure A.4: Agriculture: origin $\times$ SITC division, foreign-MNE share of export value.

Table A.7: Numbers behind the previous figure

Sub-sector	ARG	CHL	COL	CRI	DOM	PER	PRY	SLV	URY	All
<i>Panel A: composition of the origin's exports in the scope, % (columns sum to 100)</i>										
08 Animal feed	29.1	4.4	0.6	1.5	0.7	20.5	16.5	1.7	1.0	17.8
05 Vegetables & fruit	7.6	43.7	5.9	48.1	20.8	36.4	0.3	7.2	2.0	15.6
04 Cereals	20.9	2.8	2.5	3.6	5.6	3.3	12.5	19.6	20.5	13.4
22 Oil seeds	9.5	0.4	0.0	0.1	0.0	0.3	38.6	0.1	20.6	8.6
07 Coffee, tea, cocoa, spices	0.9	0.7	45.4	9.9	11.1	14.5	0.1	21.8	0.3	7.9
42 Fixed vegetable oils	12.8	0.5	6.2	5.2	1.2	0.5	9.1	0.2	0.1	7.7
01 Meat	5.9	9.2	1.4	2.8	0.7	0.1	18.3	1.3	30.2	7.0
03 Fish & seafood	3.9	10.4	1.6	1.8	0.6	13.1	0.0	9.6	3.0	4.9
29 Crude animal/veg. materials	0.4	3.8	23.4	3.7	1.0	2.1	1.1	1.6	1.5	3.6
11 Beverages	2.5	16.4	0.4	1.7	6.8	0.5	0.0	9.2	0.5	3.4
02 Dairy & eggs	2.8	1.4	0.3	3.9	0.6	1.5	0.2	2.4	12.7	2.6
09 Misc. edible products	1.3	3.5	2.2	15.1	6.4	0.9	0.2	5.3	2.4	2.4
06 Sugar & honey	1.1	0.5	8.8	1.1	6.7	0.8	1.1	12.1	0.6	2.1
12 Tobacco	0.9	0.8	0.6	0.3	37.1	0.1	0.8	0.0	1.1	1.5
41 Animal oils & fats	0.0	1.0	0.0	0.0	0.0	4.4	0.3	0.0	1.3	0.7
Other	0.5	0.5	0.7	0.9	0.7	1.1	0.7	7.7	2.2	0.9
All sub-sectors	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
<i>Panel B: foreign-MNE share of export value within the cell, %</i>										
08 Animal feed	30.9	24.2	79.1	58.9	91.5	35.7	86.6	24.1	39.6	35.7
05 Vegetables & fruit	29.6	26.5	17.9	44.0	14.3	40.0	45.0	38.0	18.1	32.6
04 Cereals	49.9	63.0	28.2	61.0	34.2	28.6	53.7	8.8	52.8	48.6
22 Oil seeds	55.8	31.8	18.8	1.2	0.8	50.5	73.1	4.8	55.5	61.6
07 Coffee, tea, cocoa, spices	35.4	85.1	36.2	45.6	6.3	12.4	4.0	20.2	13.7	30.4
42 Fixed vegetable oils	25.2	17.1	38.6	0.0	10.2	12.8	85.2	14.4	56.6	30.8
01 Meat	20.8	16.3	50.8	19.3	1.4	16.3	38.6	42.4	66.0	34.8
03 Fish & seafood	55.4	30.4	42.4	27.4	7.5	32.0	0.0	85.4	19.6	41.9
29 Crude animal/veg. materials	29.7	43.9	20.6	19.7	49.2	47.8	50.5	69.3	50.2	27.3
11 Beverages	27.3	20.7	31.0	22.9	63.0	27.7	48.3	87.3	42.4	28.6
02 Dairy & eggs	29.8	58.4	11.7	9.6	54.9	77.3	0.8	0.8	13.1	28.0
09 Misc. edible products	64.5	88.2	45.8	84.6	55.7	37.1	1.1	44.7	93.6	70.9
06 Sugar & honey	26.7	47.4	19.2	4.9	22.1	16.2	0.1	0.1	69.8	19.3
12 Tobacco	42.3	99.8	27.4	1.0	65.0	35.6	20.0	5.0	21.5	54.8
41 Animal oils & fats	26.3	13.3	41.6	36.8	89.4	24.9	28.7	49.8	20.5	23.0
Other	36.1	45.3	16.5	26.3	5.1	7.5	1.6	0.1	34.5	21.6
All sub-sectors	37.2	30.3	30.4	44.4	40.7	33.0	65.0	29.6	50.2	38.5

Pooled 2006–2022. Sub-sectors on the rows, exporting countries on the columns. Panel A: share of each sub-sector in the origin's exports within the scope (columns sum to 100); 'All' = the sub-sector's share of the scope's exports. Panel B: foreign-MNE share of export value in each origin × sub-sector cell ('-' = no exports); 'All' = the sub-sector's overall foreign-MNE share; last row = the origin's overall foreign-MNE share in the scope.

Table A.8: Agriculture by SITC division: parent-country composition of foreign-MNE exports

SITC Rev. 3 division	USA	CHE	NLD	LIE	BRA	ESP	DEU	SAU	LUX	GBR	Other	Domestic
08 Animal feed	6.3	9.8	4.7	6.5	0.3	0.5	0.2	0.1	0.0	1.2	6.2	11.5
05 Vegetables & fruit	3.2	0.2	0.7	0.0	0.0	3.3	1.2	0.1	0.8	0.4	22.7	5.8
04 Cereals	15.0	6.2	6.2	5.8	1.3	0.5	1.0	0.0	3.5	1.0	8.2	8.2
22 Oil seeds	30.7	3.2	7.8	4.8	3.1	0.1	0.1	0.0	0.0	0.3	11.5	4.1
07 Coffee, tea, cocoa, spices	1.5	0.8	0.0	0.0	1.2	0.1	10.2	0.0	1.1	0.0	15.6	15.6
42 Fixed vegetable oils	6.3	7.3	6.7	4.0	0.3	0.1	0.0	0.0	0.1	0.4	5.6	10.2
01 Meat	0.2	0.0	0.0	0.0	7.7	0.1	0.0	13.4	0.0	0.9	12.5	13.1
03 Fish & seafood	1.7	0.0	0.1	0.0	0.0	10.1	0.0	0.4	3.4	0.7	25.4	9.5
29 Crude animal/veg. materials	0.9	0.1	0.7	0.7	1.0	0.3	1.8	0.7	0.2	0.5	20.4	9.0
11 Beverages	0.7	0.6	0.8	0.1	0.0	1.2	0.1	0.0	3.6	2.6	19.0	9.2
02 Dairy & eggs	0.8	5.2	5.3	0.9	0.1	0.1	0.0	0.0	0.0	0.1	15.6	20.3
09 Misc. edible products	31.8	12.0	1.0	0.0	0.1	0.7	0.5	0.0	0.7	4.5	19.7	2.3
06 Sugar & honey	7.4	0.0	0.6	1.5	0.0	0.1	0.7	0.0	0.0	0.1	8.8	14.7
12 Tobacco	19.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.4	29.2	0.2
41 Animal oils & fats	0.0	0.0	1.7	0.0	0.4	0.0	0.0	1.4	0.0	3.3	16.2	11.0
Other	3.2	3.4	0.3	0.8	0.1	0.6	1.6	0.3	0.5	1.3	9.7	1.9

Share of the row's export value by parent-country group, percent. The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and 'Other foreign' (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country. The remainder of each row is local (unmatched) firms.

Table A.9: Agriculture: HS sections by export value, MNE shares and leading parent

HS section	\$bn/yr	For.	Dom.	Local	Lead parent
II Vegetable products	33.5	42	7	51	USA (0.33)
IV Prepared foodstuffs, beverages, tobacco	25.9	37	10	53	USA (0.21)
I Live animals; animal products	12.5	34	14	51	SAU (0.19)
III Fats and oils	6.9	31	10	59	CHE (0.26)
All sections	78.9	38	9	52	

HS sections ranked by export value. For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the section's export value, percent; Lead parent = largest parent country among the section's foreign MNEs and its share of the section's foreign-MNE value. Dollar values are annual averages: each origin's pooled value divided by the number of years it is observed (ARG 2011–2019, CHL 2009–2022, COL 2010–2021, CRI 2010–2019, DOM 2012–2019, PER 2010–2019, PRY 2012–2020, SLV 2006–2018, URY 2010–2019); shares are computed on the pooled values.

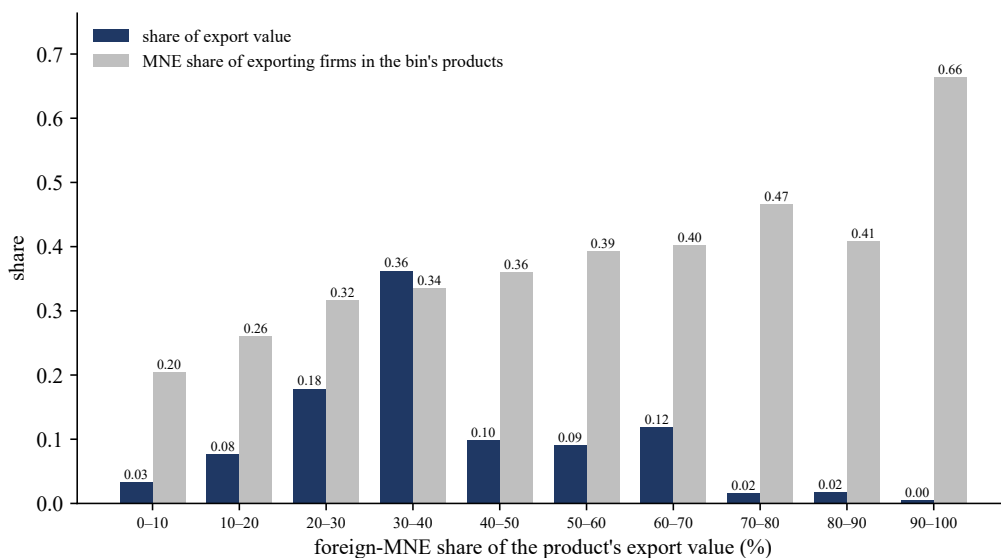
**Figure A.5:** Agriculture: distribution of the HS6 foreign-MNE share.

Table A.10: Numbers behind the previous figure

Foreign share bin (%)	Share of value	MNE share of exporting firms	N HS6
0–10	0.034	0.205	349
10–20	0.077	0.261	159
20–30	0.179	0.316	134
30–40	0.363	0.335	103
40–50	0.099	0.361	60
50–60	0.091	0.393	53
60–70	0.119	0.402	49
70–80	0.016	0.466	37
80–90	0.018	0.409	30
90–100	0.005	0.664	49

MNE share of exporting firms = matched firm-cells (firm x destination x HS6 x year) over all firm-cells in the bin's products.